nature portfolio

Peer Review File

Artificial Intelligence-derived Photoplethysmography Age as a Digital Biomarker for Cardiovascular Health

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Healthy vascular aging shapes the path to longevity, yet very few succeeded during the life course. For the identification of early vascular aging, a wearable device that registers photoplethysmography (PPG) may offer a hand. The manuscript by Nie and colleagues entitled "Artificial intelligence-derived vascular age from photoplethysmography: A novel digital biomarker for cardiovascular health", presents a novel machine-learning approach to estimate a "vascular age" metric using photoplethysmography (PPG) and chronological age data from a relatively healthy population. The study further evaluates this metric's ability to identify individuals at high risk for major adverse cardiac and cerebrovascular events (MACCE) and predict in-hospital mortality in external cohorts.

The strengths of the paper include its integration of machine learning to PPG signal analysis, the use of large datasets, a well-structured internal and external validation process, and validation using prospectively collected data. These factors contribute to extending the potential utility of PPG signals. However, from a clinical perspective, several points warrant clarification and discussion.

1. The term "vascular age" is used yet the model is trained to predict chronological age from a relatively healthy population, rather than directly modeling established markers (e.g., pulse wave velocity, augmentation index, carotid intima-media thickness). While PPG signals inherently reflect vascular properties, this specific definition of "vascular age" needs to be explicitly and clearly stated in the abstract, introduction, and methods.

2. The correlation between the predicted "vascular age" and actual chronological age is reported as moderate (e.g., Pearson r=0.49 in the UK Biobank holdout set, Table 2). While it might be sufficient for identifying high-risk groups, it indicates limitations in precision and should be discussed as a limitation. Of note, the manuscript currently lacks a dedicated "Limitations" section, which is to be added.

3. PPG is significantly influenced by blood pressure levels rather than just binary hypertension status, and the use of vasoactive medications (e.g., antihypertensives, vasodilators). Blood pressure measurements should be adjusted for during the development of the "vascular age" model within the healthy training cohort. Additionally, the potential confounding effects of relevant medications appear not to have been incorporated. These factors should be addressed, and their absence in the model development may reduce performance in external dataset with medications. This should be acknowledged as a limitation.

4. In the multivariable analyses evaluating the association of the "vascular age gap" with MACCE and mortality, established cardiovascular risk factors, especially continuous variables, should be accounted, similar to those used in standard risk scores (e.g., Framingham, SCORE2, QR4). Systolic blood pressure should be included, due to its relationship with PPG signals and future cardiovascular risk. If the "vascular age gap" remains significantly associated with outcomes after adjusting for SBP and other cardiovascular covariates, the authors should evaluate the added predictive value (C-statistic, NRI, IDI).

5. The concept of a "vascular age gap" is introduced, defined as a predicted age exceeding chronological age by more than one standard deviation (+1 SD). However, the absolute value corresponding to +1 SD varies across the different datasets (e.g., 9 years vs. 15 years). The authors should consider either: (a) exploring the use of standardized, potentially rounded,

absolute thresholds (e.g., +5 or +10 years) based on clinical relevance or optimal risk prediction cut-points, or (b) explicitly discussing the limitation that the +1 SD threshold lacks consistent interpretation across populations. Furthermore, the stability of this gap metric, particularly in individuals prone to significant hemodynamic changes (e.g., acutely ill patients), warrants discussion. Without population-specific calibration or further validation in dynamic clinical states, the generalizability and stability of the currently defined gap may be limited.

Reviewer #2

(Remarks to the Author)
Summary:
This manuscript presents a compelling and timely contribution to the growing field of digital biomarkers by introducing a deep learning model that estimates vascular age from PPG signals. The authors propose a novel loss function to address imbalanced regression and validate their approach across three large, diverse datasets. Overall, this work has strong clinical relevance and considerable translational potential. The concept of a wearable-derived vascular age biomarker is particularly appealing given the increasing adoption of consumer health devices.
I do have a few technical comments and suggestions that I believe would improve clarity, reproducibility, and interpretability of the work.

Major Comments:

1. Model performance and clinical use case
While the vascular age gap is shown to be predictive of adverse cardiometabolic outcomes, the reported MAE (7 to 12 years across datasets) and moderate correlation values (r = 0.38 - 0.54) raise the question of how this biomarker would be interpreted in real-world clinical settings. I feel it would be better to include a discussion contextualizing this level of error. For example, how large a gap is considered ‘clinically meaningful’ given this baseline model variability? Could false positives or negatives have unintended consequences if this tool is deployed broadly?

2. Transparency of implementation details
The methodology section provides a good overview of the model architecture and training setup. However, important details are either in the supplementary material or left unspecified:
- What was the total number of learnable parameters in the Net1D model?
- Were any other models evaluated before zeroing on Net1D (e.g., GRU, LSTM, or transformer-based models for sequential PPG data)?
- Was any data augmentation or signal denoising performed, especially for the noisier ICU datasets?

3. Validation strategy across domains
There are substantial differences between the PPG data formats in UK Biobank (single-beat averaged waveforms) and PulseDB (10 second signals). While fine-tuning was performed on PulseDB VitalDB, it is unclear how the model generalizes to entirely different acquisition conditions. Can the authors comment on how signal duration and format impact performance? Is there any evidence that the model trained on one format degrades when applied to the other?

4. Subgroup and bias analysis
The authors demonstrate associations between vascular age gap and key outcomes, but do not report stratified analyses across sex, ethnicity, or prevalent disease status (beyond initial exclusions). Given well-documented disparities in vascular aging and health device accuracy across populations, this is a noticeable omission. It would be better to understand if model bias exists and whether predictive accuracy differs across demographic subgroups.

5. Interpretability and saliency maps
The inclusion of saliency maps is much appreciated. However, the methods used for saliency computation are not clearly described. Were gradients or integrated gradients used? Was any smoothing or averaging applied? A more detailed explanation would help readers assess the robustness of these visualizations.

Minor Comments:

1. Dist loss clarification:
The Dist Loss formulation is interesting and seems appropriate for the skewed age distribution in the UKB dataset. However, it would help to include a mathematical description or pseudocode in the supplement (Figure S1 is helpful but lacks detail on the actual KDE computation). Additionally, it would be good to clarify whether this loss has been benchmarked against other strategies like focal loss or re-weighted regression losses.

2. Typographical and terminology inconsistencies
- Some of the figures (e.g spline curves and KM curves) use small fonts, which may be difficult to interpret. Consider making these more accessible.
- The clinical endpoints from UKB and MIMIC should be consolidated into a single supplementary table for easier cross-reference, especially as ICD-9/10 codes and outcome definitions differ between cohorts.
- A very minor detail, but if the neural network schematic in figure 1 (a) could be made vertical, it would be better.
- Font size in Figure 2 is extremely small. Please consider revising it.

3. Future directions
The discussion could be strengthened by outlining a roadmap for clinical translation. Could this model be embedded into smartwatches or pulse oximeters? Would longitudinal monitoring improve predictive power? These additions would help readers envision the practical impact of this work.

Overall impression
This is a strong manuscript with broad relevance to digital health, cardiovascular aging, and AI-driven biomarker development. The methods are mostly rigorous, the findings are clinically meaningful, and the innovation is clear. Addressing the concerns noted above, particularly around transparency, subgroup performance, and practical error interpretation, will further strengthen the impact of the paper. Great work!

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
All comments addressed. No further comments.

Reviewer #2

(Remarks to the Author)
Thank you for your thorough revision of the manuscript. The authors have addressed all of my previous comments and questions in a clear manner. The additional explanations and modifications have strengthened the clarity, methodology, and overall presentation of the work. I find the revised manuscript to be well prepared, and I have no further concerns.

Dear Editor and Reviewers,

We would like to express our sincere gratitude for your time, thoughtful comments, and constructive suggestions on our manuscript. We truly appreciate your efforts, which have significantly contributed to improving the clarity, rigor, and overall quality of our work.

In the following, we provide detailed, point-by-point responses to all the comments raised. For each comment, we outline our response and indicate the corresponding revisions made to the manuscript. All changes in the revised version are highlighted in blue for ease of reference.

Once again, we are sincerely grateful for your insightful feedback and the opportunity to improve our work.

Reviewer #1

Comment 1: The term "vascular age" is used yet the model is trained to predict chronological age from a relatively healthy population, rather than directly modeling established markers (e.g., pulse wave velocity, augmentation index, carotid intima-media thickness). While PPG signals inherently reflect vascular properties, this specific definition of "vascular age" needs to be explicitly and clearly stated in the abstract, introduction, and methods.

Response: We sincerely thank the reviewer for the insightful and important comment regarding the terminology of "vascular age". This observation highlights a critical need for precise definition to avoid potential misunderstandings. In response, we have replaced the term "vascular age" with "AI-PPG age" throughout the manuscript to better reflect that this digital biomarker is derived from AI-based analysis of PPG signals, rather than directly measuring traditional vascular aging indices such as pulse wave velocity. Furthermore, we have explicitly clarified this concept in the manuscript to improve conceptual clarity and help readers accurately interpret the scope and nature of our model's predictions. We believe these revisions strengthen the manuscript's precision and enhance its contribution to the field.

Comment 2: The correlation between the predicted "vascular age" and actual chronological age is reported as moderate (e.g., Pearson r=0.49 in the UK Biobank holdout set, Table 2). While it might be sufficient for identifying high-risk groups, it indicates limitations in precision and should be discussed as a limitation. Of note, the manuscript currently lacks a dedicated "Limitations" section, which is to be added.

Response: We thank the reviewer for the valuable suggestion regarding the moderate correlation between AI-PPG age and chronological age. In the revised manuscript, we have added a dedicated "Limitations" section in the Discussion (Page 13, Paragraph 6, Lines 1-6), where this issue is explicitly acknowledged and discussed.

This study has several limitations. First, the results demonstrated a moderate correlation between AI-PPG age and calender age, which may be attributed to the limited age-related information captured in the relatively short PPG recordings. While this does not compromise the utility of AI-PPG age in cardiovascular risk stratification, improving its numerical alignment with calender age could enhance interpretability in practical applications. One potential approach is to apply post hoc correction strategies, such as the bias adjustment methods proposed in previous studies Beheshti et al. [2019], Le Goallec et al. [2022], Le et al. [2018]. Second, the waveform of PPG signals is not only affected by

While the model exhibits a moderate correlation (e.g., Pearson r = 0.49 in the UK Biobank holdout set), this level of agreement is expected given the inter-individual heterogeneity in vascular aging and the limited temporal information contained in short PPG segments. Importantly, our aim is not to precisely estimate chronological age but rather to provide an AI-derived biomarker that reflects vascular health. As demonstrated by our outcome analyses, the deviation between AI-PPG age and chronological age is predictive of adverse cardiovascular events, highlighting the utility of the model for risk stratification despite its moderate precision. We hope this clarification adequately addresses the reviewer's concern.

Comment 3: PPG is significantly influenced by blood pressure levels rather than just binary hypertension status, and the use of vasoactive medications (e.g., antihypertensives, vasodilators). Blood pressure measurements should be adjusted for during the development of the "vascular age" model within the healthy training cohort. Additionally, the potential confounding effects of relevant medications appear not to have been incorporated. These factors should be addressed, and their absence in the model development may reduce performance in external dataset with medications. This should be acknowledged as a limitation.

Response: We thank the reviewer for raising this important point regarding potential confounding from blood pressure levels and medication use. This observation highlights a key consideration for developing physiologically grounded and generalizable models.

As noted by the reviewer, PPG signals are indeed influenced by both blood pressure and vasoactive medications. However, we intentionally chose not to include these variables during model development, as our primary goal was to build a practical and deployable biomarker using only raw PPG signals, enabling use in wearable devices without requiring additional clinical inputs. We believe this design choice supports scalability and simplicity for population-level screening.

That said, we acknowledge that this decision introduces limitations, particularly when applying the model to populations with active pharmacologic treatment.To address this, we have explicitly added a statement acknowledging the potential impact of unmodeled variables such as blood pressure and medication use in the Discussion section (Page 13, Paragraph 6, continued on Page 14, Lines 6-11). We appreciate the reviewer's suggestion, which has helped us better articulate this trade-off.

et al. [2019], Le Goallec et al. [2022], Le et al. [2018]. Second, the waveform of PPG signals is not only affected by vascular aging but also by other physiological factors such as blood pressure and the use of vasoactive medications. In this study, to ensure ease of use and convenience, we developed the AI-PPG age prediction model using only raw PPG signals without incorporating additional covariates. While this design improves applicability, it may limit the model's ability to comprehensively capture cardiovascular health. Future work may explore combining AI-PPG

age with other cardiovascular-related indicators at the application stage to further improve assessment performance.

Comment 4: In the multivariable analyses evaluating the association of the "vascular age gap" with MACCE and mortality, established cardiovascular risk factors, especially continuous variables, should be accounted, similar to those used in standard risk scores (e.g., Framingham, SCORE2, QR4). Systolic blood pressure should be included, due to its relationship with PPG signals and future cardiovascular risk. If the "vascular age gap" remains significantly associated with outcomes after adjusting for SBP and other cardiovascular covariates, the authors should evaluate the added predictive value (C-statistic, NRI, IDI).

Response: We thank the reviewer for this insightful and detailed suggestion regarding the multivariable analyses. We agree that incorporating established cardiovascular risk factors, especially continuous variables such as systolic blood pressure (SBP), is crucial for rigorously evaluating the independent association of the vascular age gap with clinical outcomes.

In response, we constructed a multivariable Cox model including clinical variables from the Framingham risk score, explicitly adjusting for SBP. As shown in Table 3 of the revised manuscript, the AI-PPG age gap remained significantly associated with both primary and secondary outcomes after controlling for these covariates.

Model 3, adjusted for age, sex, SBP, use of antihypertensive treatment, smoking status, diabetes, total cholesterol, and HDL cholesterol.						
MACCE	1.02 (1.02-1.02)	<0.005	0.84 (0.81-0.88)	<0.005	2.37 (2.14-2.62)	<0.005
Coronary heart disease	1.03 (1.03-1.04)	<0.005	0.74 (0.71-0.78)	<0.005	2.33 (2.10-2.59)	<0.005
Heart failure	1.02 (1.01-1.02)	<0.005	0.86 (0.80-0.92)	<0.005	2.41 (2.01-2.89)	<0.005
Myocardial infraction	1.04 (1.03-1.04)	<0.005	0.71 (0.66-0.77)	<0.005	2.62 (2.23-3.08)	<0.005
Stroke	1.02 (1.02-1.03)	<0.005	0.85 (0.77-0.93)	<0.005	2.33 (1.87-2.91)	<0.005
All-cause mortality	1.01 (1.01-1.02)	<0.005	0.94 (0.89-0.99)	0.02	2.32 (2.02-2.66)	<0.005

Furthermore, we evaluated the incremental predictive value of the AI-PPG age gap. Results shown in Supplementary Table S3 indicate that it improves discrimination and calibration beyond existing clinical risk scores, highlighting its potential as an AI-derived biomarker for cardiovascular risk assessment.

Comment 5: The concept of a "vascular age gap" is introduced, defined as a predicted age exceeding chronological age by more than one standard deviation (+1 SD). However, the absolute value corresponding to +1 SD varies across the different datasets (e.g., 9 years vs. 15 years). The authors should consider either: (a) exploring the use of standardized, potentially rounded, absolute thresholds (e.g., +5 or +10 years) based on clinical relevance or optimal risk prediction cut-points, or (b) explicitly discussing the limitation that the +1 SD threshold lacks consistent interpretation across populations. Furthermore, the stability of this gap metric, particularly in individuals prone to significant hemodynamic changes (e.g., acutely ill patients), warrants discussion. Without population-specific calibration or further validation in dynamic clinical states, the generalizability and stability of the currently defined gap may be limited.

Response: We appreciate the reviewer's thoughtful comments regarding the threshold definition of the AI-PPG age gap metric. In this study, we adopted the +1 standard deviation (SD) threshold following precedents in prior work[1]**Error! Reference source not found.Error! Reference source not found.Error! Reference source not found.**. As the reviewer rightly points out, the absolute value corresponding to 1 SD varies across datasets: approximately 9 years in the general population versus 15 years in critically ill patients. This reflects the increased variability in AI-PPG age under hemodynamic instability, where a wider threshold may better capture high-risk individuals.

We acknowledge that using a +1 SD threshold introduces challenges across different populations. To address this, we have added a discussion in the Limitations section (Page 13, Paragraph 6, continued on Page 14, Lines 12-16) highlighting that the threshold may require calibration depending on specific clinical contexts. We also agree that further validation, particularly in dynamically changing physiological states, is essential to ensure the robustness and generalizability of the vascular age gap metric. We thank the reviewer for raising this important point, which prompted us to clarify the applicability and limitations of our thresholding approach more explicitly in the revised manuscript.

Third, following previous studies Al-Falahi et al. [2025], Cho et al. [2025], Lima et al. [2021], we used the standard deviation of the AI-PPG age gap as a threshold for stratifying individuals. However, this data-driven threshold may vary across populations, raising concerns about its generalizability. While our findings in the UKB cohort support a 9-year threshold for the general population, further validation is needed before applying this cutoff in other demographic or clinical settings. Finally, AI-PPG age is intended to serve as an auxiliary indicator of cardiovascular risk rather than a

Reviewer #2

Comment 1: Model performance and clinical use case. While the vascular age gap is shown to be predictive of adverse cardiometabolic outcomes, the reported MAE (7 to 12 years across datasets) and moderate correlation values (r = 0.38 - 0.54) raise the question of how this biomarker would be interpreted in real-world clinical settings. I feel it would be better to include a discussion contextualizing this level of error. For example, how large a gap is considered 'clinically meaningful' given this baseline model variability? Could false positives or negatives have unintended consequences if this tool is deployed broadly?

Response: We sincerely thank the reviewer for raising this important point regarding the interpretability of model error and its implications for real-world clinical use. We agree that the reported MAE and moderate correlation warrant careful contextualization to clarify the practical significance of the vascular age gap as a biomarker.

In the revised manuscript, we have added discussion to address these concerns from three perspectives (Page 13, Paragraph 6, continued on Page 14, Lines 1-6, 12-16): First, regarding clinical interpretability, we note that the threshold for a "meaningful" age gap may vary depending on the population and use case. For instance, prior analyses in the UKB cohort suggest that a gap of 9 years can effectively stratify cardiovascular risk in the general population. However, in settings with greater physiological instability (e.g., ICU patients), larger fluctuations are expected, and the interpretive threshold may need adjustment accordingly. Second, we acknowledge that false positives or negatives could have unintended consequences, especially if used in isolation. To mitigate this, we emphasize that our vascular age metric is intended as an auxiliary indicator, and not a standalone diagnostic tool. We have clarified in the revision that it should be interpreted in conjunction with other clinical assessments to support rather than replace medical decision-making (Page 13, Paragraph 6, continued on Page 14, Lines 16-18). Finally, despite the prediction error, we believe that the non-invasive and scalable nature of PPG-based age estimation offers potential value as a population-level screening tool, particularly in resource-limited settings. Its utility lies in broad risk stratification and longitudinal health monitoring, rather than precise individual-level diagnosis. We hope these clarifications help contextualize the error metrics and address the reviewer's thoughtful concerns.

This study has several limitations. First, the results demonstrated a moderate correlation between AI-PPG age and calender age, which may be attributed to the limited age-related information captured in the relatively short PPG recordings. While this does not compromise the utility of AI-PPG age in cardiovascular risk stratification, improving its numerical alignment with calender age could enhance interpretability in practical applications. One potential approach is to apply post hoc correction strategies, such as the bias adjustment methods proposed in previous studies Beheshti et al. [2019], Le Goallec et al. [2022], Le et al. [2018]. Second, the waveform of PPG signals is not only affected by

Third, following previous studies Al-Falahi et al. [2025], Cho et al. [2025], Lima et al. [2021], we used the standard deviation of the AI-PPG age gap as a threshold for stratifying individuals. However, this data-driven threshold may vary across populations, raising concerns about its generalizability. While our findings in the UKB cohort support a 9-year threshold for the general population, further validation is needed before applying this cutoff in other demographic or clinical settings. Finally, AI-PPG age is intended to serve as an auxiliary indicator of cardiovascular risk rather than a standalone diagnostic tool. Its use in real-world settings should be integrated with other clinical information to avoid over-intervention due to outliers or underestimation of risk due to occasional prediction errors.

Comment 2: Transparency of implementation details. The methodology section provides a good overview of the model architecture and training setup. However, important details are either in the supplementary material or left unspecified:
- What was the total number of learnable parameters in the Net1D model?
- Were any other models evaluated before zeroing on Net1D (e.g., GRU, LSTM, or transformer-based models for sequential PPG data)?
- Was any data augmentation or signal denoising performed, especially for the noisier ICU datasets?

Response: We appreciate the reviewer's insightful comments regarding the transparency of implementation details. To improve clarity and reproducibility, we have addressed the following points:

1. Model size: The Net1D model used in our experiments contains 389,689 trainable parameters. This information has now been included in the revised manuscript (Page 4, Section 2.2.1, Lines 7-8).

   of PPG signals for AI-PPG age prediction. In our implementation, the Net1D-based model comprises 389,689 learnable parameters, and the detailed model configuration and pretrained weights are provided.

2. Model selection rationale: Based on prior literature on PPG-based modeling**Error! Reference source not found.Error! Reference source not found.Error! Reference source not found.**, CNN-based architectures are commonly used due to their computational efficiency and suitability for capturing local waveform morphology. We also conducted preliminary experiments comparing Net1D to GRU, LSTM, and Transformer variants, and found that CNNs consistently outperformed these alternatives in both convergence and final accuracy.

3. Data augmentation and denoising: As mentioned in Section 2.1, we did not apply additional augmentation or denoising strategies. The raw signals from both UK Biobank and PulseDB were of sufficient quality. Only Z-score normalization was performed during preprocessing.

We hope these clarifications address the reviewer's concerns and improve the transparency and reproducibility of our methodology.

Comment 3: Validation strategy across domains. There are substantial differences between the PPG data formats in UK Biobank (single-beat averaged waveforms) and PulseDB (10 second signals). While fine-tuning was performed on PulseDB VitalDB, it is unclear how the model generalizes to entirely different acquisition conditions. Can the authors comment on how signal duration and format impact performance? Is there any evidence that the model trained on one format degrades when applied to the other?

Response: We sincerely appreciate the reviewer's insightful comment regarding the differences in PPG data formats between the UK Biobank and PulseDB datasets. This observation highlights a critical consideration for model generalizability. Specifically, the UK Biobank provides single-beat averaged waveforms, whereas PulseDB contains continuous 10-second recordings. Such discrepancies in signal duration and structure may introduce challenges when transferring models across domains.

In particular, inputs containing only a single beat may lack important inter-beat temporal context, such as rhythm or beat-to-beat variability, which can affect the model's ability to capture global waveform dynamics. These differences may lead to distribution shifts that challenge the robustness of the model across datasets.

To address this issue, we adopted a fine-tuning strategy whereby the model initially trained on UK Biobank data was further fine-tuned on the VitalDB subset of PulseDB, which features continuous waveform data more closely resembling the external validation setting. This adaptation enhances the model's compatibility with varying acquisition conditions.

Comment 4: Subgroup and bias analysis. The authors demonstrate associations between vascular age gap and key outcomes, but do not report stratified analyses across sex, ethnicity, or prevalent disease status (beyond initial exclusions). Given well-documented disparities in vascular aging and health device accuracy across populations, this is a noticeable omission. It would be better to understand if model bias exists and whether predictive accuracy differs across demographic subgroups.

Response: We thank the reviewer for raising the important point regarding subgroup and bias analyses. In our multivariable Cox regression models, we have already adjusted for key demographic and clinical variables such as sex, ethnicity, and prevalent disease status, and the vascular age gap remained a significant predictor after these adjustments. This suggests that the model's predictive value is robust across these factors.

Furthermore, we have expanded the subgroup analyses in Supplementary Figure S3, presenting results stratified by age, sex, and prevalent conditions including hypertension, diabetes, dyslipidemia, and chronic kidney disease. These subgroup results consistently demonstrate the significant association of the vascular age gap across diverse demographic and clinical strata, indicating stable predictive performance among different subpopulations. Together, these findings support the broad applicability of the vascular age gap as a predictive biomarker across varied population groups.

Comment 5: Interpretability and saliency maps. The inclusion of saliency maps is much appreciated. However, the methods used for saliency computation are not clearly described. Were gradients or integrated gradients used? Was any smoothing or averaging applied? A more detailed explanation would help readers assess the robustness of these visualizations.

Response: We thank the reviewer for the careful attention to the saliency map analysis. As detailed in the manuscript (Page 5, Section 2.2.4), we employed a standard gradient-based approach (i.e., raw gradients of the output with respect to input) to compute the saliency maps, rather than integrated gradients or other variants. To enhance visualization clarity and reduce noise, the raw saliency maps were smoothed using a Gaussian filter. Furthermore, to improve robustness and mitigate individual variability, the presented saliency maps represent the average overlay across all subjects at a given age (e.g., 40 years), instead of single-subject examples. This averaging helps highlight consistent model attention patterns rather than idiosyncratic noise. We hope this explanation addresses the reviewer's concerns and clarifies the robustness of our visualizations.

Minor comment 1: Dist loss clarification. The Dist Loss formulation is interesting and seems appropriate for the skewed age distribution in the UKB dataset. However, it would help to include a mathematical description or pseudocode in the supplement (Figure S1 is helpful but lacks detail on the actual KDE computation). Additionally, it would be good to clarify whether this loss has been benchmarked against other strategies like focal loss or re-weighted regression losses.

Response: We sincerely appreciate the reviewer's insightful comments and attention to the Dist Loss. To enhance clarity, we have added detailed pseudocode of the Dist Loss computation in Supplementary Figure S2, which we believe will assist readers in better understanding the algorithmic steps involved. Furthermore, in our previous work, we have demonstrated the superior performance of Dist Loss compared to cost-sensitive learning and other imbalanced regression methods across multiple datasets[Error! Reference source not found.]. We hope that these additions and clarifications satisfactorily address the reviewer's concerns and highlight the effectiveness of the proposed loss function.

Minor comment 2: Typographical and terminology inconsistencies.

- Some of the figures (e.g spline curves and KM curves) use small fonts, which may be difficult to interpret. Consider making these more accessible.
- The clinical endpoints from UKB and MIMIC should be consolidated into a single supplementary table for easier cross-reference, especially as ICD-9/10 codes and outcome definitions differ between cohorts.
- A very minor detail, but if the neural network schematic in figure 1 (a) could be made vertical, it would be better.
- Font size in Figure 2 is extremely small. Please consider revising it.

Response: We sincerely thank the reviewer for the valuable and constructive suggestions regarding presentation and terminology consistency. In response, we have increased the font sizes of the spline curves, KM curves, and Figure 2 to enhance readability. The neural network schematic in Figure 1(a) has also been revised to a vertical layout as recommended.

Regarding the suggestion to consolidate clinical endpoints from UKB and MIMIC into a single supplementary table, we acknowledge the intent to facilitate cross-reference between cohorts. However, given the inherent differences in ICD-9/10 coding systems and outcome definitions across these datasets, the existing separate tables provide clear and detailed cohort-specific information, thereby preserving clarity and avoiding confusion. We believe this organization best supports accurate interpretation of the endpoints.

We appreciate the reviewer's detailed feedback, which has contributed significantly to improving the clarity and quality of our manuscript.

Minor comment 3: Future directions. The discussion could be strengthened by outlining a roadmap for clinical translation. Could this model be embedded into smartwatches or pulse oximeters? Would longitudinal monitoring improve predictive power? These additions would help readers envision the practical impact of this work.

Response: We thank the reviewer for this valuable suggestion. We agree that outlining potential pathways for clinical translation strengthens the discussion. Accordingly, the revised manuscript includes a section addressing the feasibility of embedding the AI-PPG age model into widely used wearable devices such as smartwatches and pulse oximeters, enabling convenient, real-time cardiovascular health monitoring. We believe this addition better contextualizes the practical significance and future clinical applications of our work.

intuitive and interpretable proxy for vascular health, promoting individual awareness and self-monitoring. Importantly, AI-PPG age is derived exclusively from raw PPG signals, enabling seamless integration into wearable platforms such as smartwatches. This approach offers two key clinical benefits: first, it supports large-scale cardiovascular screening and risk stratification through wearable devices; second, it facilitates targeted clinical follow-up and personalized management for individuals identified as high risk. By enabling early detection of vascular health changes and guiding timely interventions, this tiered strategy holds significant potential to improve cardiovascular outcomes.

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